Devops Assignment Day30

Task 1 — Cloud Adoption Reasoning

1. Why should the organization move to the cloud?

The organization should move to the cloud because cloud platforms provide:

- Better scalability during peak loads
- Reduced infrastructure management
- Faster deployment of applications
- Pay-as-you-use cost model

2. Which business problems will the cloud solve?

Cloud can solve:

- High infrastructure cost → no need to maintain physical servers
- Limited scalability → resources can scale automatically
- Delayed deployments → faster releases using cloud tools
- Manual monitoring → automated monitoring tools
- Fragmented authentication → centralized identity systems
- Lack of analytics → built-in analytics tools

3. Which problem will cloud NOT automatically solve?

Cloud will NOT automatically solve:

- Poor application design
- Bad database structure
- Security misconfigurations

Task 2 — Why Azure Selection Analysis

Azure is suitable for this organization.

Reasons:

- **Integration**
  Azure integrates well with Microsoft tools like Windows Server, Active Directory, and .NET applications.

- **Security**
  Provides built-in security features like identity management and access control.
- **Scalability**
  Resources can be scaled up/down based on demand.
- **Developer Ecosystem**
  Supports multiple languages and tools like Visual Studio.
- **Hybrid Support**
  Azure supports both cloud and on-premise systems (useful for gradual migration).

Task 3 — Role-Based Cloud Usage Mapping

Role	How they use	Business Benefit
Developers	Build and deploy applications	Faster Development and releases
IT Admins	Manage infrastructure and monitoring	Reduced manual work
Security Team	Manage access and security policies	Improved data protection
Executives	View Reports and Dashboards	Better decision making
Data Team	Analyze data using cloud tools	Insights from data.

Task 4 — Architecture Decision Scenario

1. High-Level Architecture Components

- Frontend (user interface)
- Backend (application logic)
- Database (store transactions)
- Authentication system
- Monitoring & alerts
- Analytics dashboard

2. Azure Service Categories

- Compute services (for backend)
- Database services
- Identity services
- Monitoring services
- Analytics services
- Networking/CDN for global access

3. Data Flow Sequence

1. User accesses portal
2. Request goes to backend
3. Authentication is verified
4. Data is stored/retrieved from database
5. Alerts triggered if needed
6. Data sent to analytics system
7. Dashboard displays insights

Task 5 — Decision Justification

Business Justification

Azure helps reduce cost, improve efficiency, and support global users.

Technical Justification

Azure provides ready-to-use services for computing, storage, and security.

Scalability Reasoning

Resources can automatically scale during high traffic, ensuring smooth performance.

Cost Reasoning

Pay-as-you-use model reduces upfront investment and avoids unused resources.

Self Evaluation

I will give a score of 13 that represent basic understanding on cloud.